Divorce-induced health decline and the impact on mortality risk*

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Abstract

Divorce is linked to elevated mortality. However, the causal mechanisms underlying this relationship are complex and remain insufficiently identified. Health is one such mechanism and is the focus of this paper. Using linked administrative data from the German Pension Insurance Fund (DRV), we examine whether divorce increases the risk of an initial musculoskeletal rehabilitation, and how much of the divorce-mortality gap runs through this channel. Rehabilitation is administered by the DRV and is granted only once a musculoskeletal condition has become chronic and threatens the ability to work. Our outcome measures the crossing of an institutional threshold rather than the onset of disease. We build annual risk sets of individuals experiencing a first (observable) divorce and individuals who have not yet divorced in that year and then use propensity score matching to generate comparable matches. Results: Divorce effect on cumulative incidence of musculoskeletal rehab, divorce effect on mortality, results from decomposition

Keywords: divorce, health, rehabilitation, mortality

JEL Classification: H75, I12, J12

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1 Introduction

The link between divorce and mortality has been widely discussed in social epidemiological studies. Register-based studies have used Finnish data (Metsä-Simola and Martikainen, 2013), Norwegian data (Berntsen and Kravdal, 2012) and English/Welsh data (Grundy and Tomassini, 2010). Until recently, Germany has been a difficult setting for this question. Brockmann and Klein (2004) is the only study to the best of our knowledge which explicitly analyzes the link between divorce and mortality although the dataset used in the study is the SOEP[1] which is not an administrative dataset. Since 2011, it is possible to link different administrative datasets from the German Pension Insurance Fund (DRV) using a unique identifier, allowing us to access full income biographies, divorce information[2], rehabilitation history and death. The availability of this dataset led to the conception of the project *Divorce and Diagnosis (DiDi): Register-based Analysis on the Role of Divorce in Health and Employment Dynamics*[3], which constitutes our study.

Our study focuses on the interaction between divorce, health and mortality. While there can be a direct link between divorce and mortality if divorce results in suicide (Kposowa, 2000; Stevenson and Wolfers, 2006) or homicide, causal evidence remains scarce when the link is indirect. Hence, we chose health as it is one possible indirect link tying together divorce and mortality. There can, however, still be threats to identification as health is not only a consequence but also a cause of divorce (Torvik et al., 2015). Furthermore, the individuals who choose to divorce may not necessarily be random. Many of the register-based studies make use of hazard models to estimate excess mortality risk after divorce and controls for sociodemographic and socioeconomic variables. For Germany, Brockmann and Klein (2004) account for health by using the self-reported health measure from the German SOEP and control for variables associated with marital status to find that divorced men have 51% higher mortality than married men while there is no significant effect for women. Leopold (2018) presents a very careful multi-outcome treatment of the gendered consequences of divorce over time, but it is not designed to control for the health that occurs before from the health consequences which follow the divorce. In our paper, we control for pre-existing health trends and other factors which occur before the divorce by using propensity score matching to answer the following questions

RQ1: Does divorce raise the risk of musculoskeletal rehabilitation[4] among divorced and non-divorced individuals?

RQ2: Is some of the excess mortality caused by divorce occurring because divorce caused additional musculoskeletal deterioration?

For context, the DRV only grants rehabilitation once the musculoskeletal condition is chronic to the point that it threatens working ability (Deutsche Rentenversicherung Bund, 2005). We do not argue that divorce causes musculoskeletal disease but rather whether divorce moves individuals who carry a latent vulnerability across the legal

[1] German Socioeconomic Panel

[2] This information is available due to the law of splitting pension points.

[3] This project is a collaboration between the Federal Institute for Population Research, Hertie School, Charité - Universitätsmedizin Berlin and the DRV.

[4] Musculoskeletal diseases are the most common cause of lost work time in Germany.

threshold where rehabilitation becomes necessary. Please see Section 2.3 for a more detailed discussion.

Results... maybe policy implications?

Section 2 provides a literature review, Section 3 introduces the data and methodology, Section 4 presents the results, Section 5 discusses the implications and Section 6 concludes.

2 Literature review

2.1 Link between divorce and mortality

The link between divorce and mortality risk is widely documented in sociology and epidemiology. In a meta-analysis, Shor et al. (2012) run meta-regressions on 625 mortality risk estimates over 104 studies and find that the mean hazard ratio was 1.30 for divorced individuals. The hazard ratio is slightly higher for men at 1.37 and lower for women at 1.22, although the difference between genders decrease with age. However, heterogeneity on the effect on death exists depending on the time since divorce. Metsä-Simola and Martikainen (2013) find that excess mortality among Finnish men, tied to accidental, violent or alcohol-related deaths, is highest immediately after divorce and declines over time. On the other hand, Berntsen and Kravdal (2012) finds that excess mortality among Norwegian men increases with time since divorce and remarriage can change the magnitude of the effect. In Germany, the evidence is limited. Brockmann and Klein (2004) use event-history models which account for marital transitions and duration in each marital status. They find evidence of selection into marriage and that mortality risk is lower when the marriage duration is longer.

Most of the previous studies establish a robust association, however, causal evidence remains scarce as divorce is not random. Sbarra and Nietert (2009) follow 1376 individuals in the Charleston Heart Study for forty years and show that the definition of the exposure determines the size of the effect. Individuals who are continuously separated or divorced have a higher hazard ratio (1.66) compared to individuals who have ever been separated or divorced (0.98). Wiborg and Gugushvili (2025) initially find raised hazard ratios for divorced men (1.59) and women (1.47) using Norwegian data. After using twin-pairs to account for genetic and early developmental and environmental factors, the relationship between divorce and mortality disappears.

2.2 Divorce and health

The effect of divorce on health is difficult to isolate due to the possibility of reverse causality. Joung et al. (1998), using Dutch data, find that married individuals who had more subjective health complaints were 1.5 times more likely to divorce, while individuals who reported multiple chronic conditions were twice as likely to divorce. Torvik et al. (2015) corroborates Joung et al. (1998)'s findings with Norwegian data. Poor health behaviors including smoking, drinking, and lack of exercise in addition to mental distress predict subsequent divorce. Metsä-Simola and Martikainen (2013), with Finnish data, find that psychotropic medication use among individuals who divorce is already elevated five years before hand and peaks at around six to nine months just

before divorce. Wade and Pevalin (2004) find worse mental health among those divorcing or divorced than among the married using British data. Generally, poor health tends to precede divorce.

To deal with health selection, Kohn and Averett (2014) estimates a dynamic model that accounts for the persistence of health and how it interacts with relationship status using 18 waves of the British Household Panel Survey. The result is that divorce has a marginally significant effect on the health of men younger than 45. Sbarra et al. (2014) also finds only a marginal effect of divorce on health. After matching individuals on divorce probability, Sbarra et al. (2014) finds that only individuals who were already depressed before divorce experienced major depressive episodes following divorce. With respect to mental health, Björkenstam et al. (2013) find that divorced individuals have thrice the risk of psychiatric inpatient care compared with married individuals in Sweden. Meulman et al. (2023) find similar results in the Netherlands where divorced individuals had higher health expenditures dedicated to mental health care. In terms of physical health, the evidence is heterogeneous. Zulkarnain and Korenman (2019) find that women who divorce at older ages tend to have more physical changes than men. Leopold (2018), on the other hand, find that men are more vulnerable in the short run in terms of well-being although the gender differences narrow in the medium run. Pellón-Elexpuru et al. (2024), in a meta-analysis across 94 studies find that divorced individuals report worse self-rated physical conditions with higher complaints of musculoskeletal, cardiovascular and cerebrovascular conditions although the effect is not significant.

2.3 Divorce and musculoskeletal disease

Evidence on the pathway from divorce to musculoskeletal impairment is rather limited and ambiguous. However, we chose to focus on musculoskeletal diseases because they are a major source of disability and work interruption in Germany[5]. The DRV Reha-Atlas (Deutsche Rentenversicherung Bund, 2024) reports that, of all completed medical rehabilitations, orthopedic conditions are the most common, accounting for 35.6% of rehabilitations among women and 35.5% among men. Brenscheidt et al. (2026) reports that musculoskeletal diseases account for 21% of certified incapacity days in 2024, which is the largest diagnostic group.

Sjøgren et al. (2009), using Danish data, find that chronic pain is more common among individuals who are divorced, separated or widowed, where musculoskeletal diseases accounted for 66.8% of the cause of chronic pain. However, Choi and Wagner (2026), using the Health and Retirement Survey find that individuals who transition from marriage to divorce tend to have a lower probability of high-impact pain. On the other hand, Cimas et al. (2026) using SHARE[6] data, find that there are gender differences. While divorced women have a higher incidence of chronic musculoskeletal pain, men have a lower prevalence. Meanwhile, Andorsen et al. (2017) finds that there is no significant effect between divorce and musculoskeletal complaints using Norwegian data.

While the impact of divorce on musculoskeletal diseases is ambiguous, the evidence on

[5]Please see the section on Rehabilitation system in Germany in the Appendix.

[6]Survey of Health, Aging and Retirement in Europe

the relationship between social support and the consequences of musculoskeletal diseases is more homogeneous. Prang et al. (2015) finds that individuals who have support from family friends and neighbors tend to have less persistent pain associated with musculoskeletal diseases, while widowed, separated or divorced individuals, in the absence of a network, tend to have a higher probability of persistent pain. Woods (2005) corroborates this and finds that a lack of support is associated with higher musculoskeletal morbidity, absence from work and subsequently, failure to return to work. Reese et al. (2010) finds that the quality of marriages matter, i.e. individuals with arthritis in problematic marriages tend to have more pain complaints and disability compared to individuals in well-adjusted marriages. This could signify that a decline in the relationship quality prior to divorce leads to poorer support, hence more complaints of pain. Couch et al. (2015) and Tamborini et al. (2016) find that divorce is associated with subsequent health threatening the ability to work, especially among individuals who do not remarry, although they do not differentiate by types of diseases.

2.4 Musculoskeletal disease and death

Musculoskeletal conditions are rarely direct causes of death in the way that cardiovascular diseases are. They often operate indirectly, for example through inflammation affecting the cardiovascular disease risk. Rheumatoid arthritis and chronic inflammatory activity can contribute to oxidative stress, dysfunction of blood vessels, arterial build-up and other cardiovascular risks (England et al., 2018). Cardiovascular mortality among people with rheumatoid arthritis is about 60% higher than the general population (Meulman et al., 2009). For non-inflammatory conditions, musculoskeletal conditions causes pain and restricted movement, which then makes people avoid physical activity. This reduces cardiovascular health and fitness, which then potentially increases the risk of comorbidities. Nüesch et al. (2011) finds that English patients with knee or hip osteoarthritis tend to have higher all-cause and cardiovascular mortality. Hawker et al. (2014) finds a similar result with Canadian patients.

Jansson, Mittendorfer-Rutz, and Alexanderson (2012) is more related to our study. Using Swedish register data, the authors compared individuals who were absent from work due to musculoskeletal diseases with people who were absent from work due to other diseases and people who were not absent from work. They find that among three categories of musculoskeletal diseases, the all-cause morbidity was 1.3 to 1.5 times higher. Given that musculoskeletal diseases account for the highest share of rehabilitations facilitated by the DRV, it is important that rehabilitation can encourage return to work and by extension, reduce the all-cause mortality attributed to musculoskeletal diseases. Bethge et al. (2019) finds that work-related medical rehabilitation tends to increase the return to work rate by 5.8 percentage points and shortens the median time to return by 9.46 days compared to conventional medical rehabilitation. In our study, the first completed DRV-facilitated musculoskeletal rehabilitation should be interpreted as an indicator of serious, work-relevant impairment and also receipt of potentially beneficial treatment.

3 Data and methodology

3.1 Data

The Research Data Center of the DRV allows linkages of different datasets starting from the reporting year 2011 through a unique identifier (Forschungsdatenzentrum Rentenversicherung, 2025). The datasets we use are AKVS[7], LTVS[8], RTBN[9], RTWF[10], RSD[11] and VA[12]. The time period covered by the data is 2011 to 2023. All estimates use the 20% sample extract, which contains 17,559,407 individuals and 200,295,283 person-years.

The AKVS, LTVS and RTBN are merged by the DRV to form the BASE dataset. The AKVS dataset contains individuals who are still actively contributing to the pension fund hence it generally represents individuals who are employed. The LTVS dataset contains individuals who have paused pension fund contributions but have not yet retired (nor drawn from their pension fund). It may include individuals who are unemployed, undergoing further education, on parental leave, etc. The RTBN dataset contains individuals who are drawing from their pension fund, i.e. retired individuals. The merged dataset, BASE, allows us to map out the full life-course coverage of individuals. If we only considered only RTBN, the sample is biased towards individuals who have contributed and lived long enough to draw a pension, ignoring those individuals who have undergone rehabilitation or died before drawing a pension. Considering only AKVS ignores rehabilitation efforts and gray divorces. To round up the dataset, it is also important to consider LTVS which details what happens between employment spells and retirement.

We then supplement the merged dataset with the RTWF, RSD and the VA. The RTWF details pension terminations due to deaths. If the death information from the merged dataset is inconsistent with the information in the RTWF, the individuals (cases) are dropped. This affects 102 individuals. Addtionally, when individuals have duplicate rows arising from BASE or RTWF, we keep the record with the fewest missing values. This affects 10,345,764 person-years with 11,297,317 surplus records. The RSD dataset details the completed rehabilitations and include information such as the broad diagnosis group, the type of rehabilitation benefits applied for, start and end date of rehabilitation. The RSD dataset also includes individuals with withdrawn, transferred or rejected applications. We only include individuals whose applications were successful and who completed the rehabilitation spell. Furthermore, about 5% of individuals undergo three or more rehabilitation spells per year. We also drop these individuals. We then supplement the merged dataset by including the additional rehabilitation information for the remaining individuals in the RSD dataset. The VA dataset includes information related to divorce and only contains divorced individuals. From the VA, we can obtain the date of divorce, pension supplements and deductions which result from divorce. Note that the VA only records divorces which have become final and where pension rights were equalized (See appendix on Updating the evolution of $\mathbf{L}_{i,t}$).

[7] Active Insured Persons
[8] Latent Insured Persons
[9] Insured Persons' Pension Portfolio
[10] Termination of Pensions due to Death
[11] Completed Rehabilitation in the Insurance History
[12] Pension Equalization Statistics

3.2 Matching

Since divorce is not randomly assigned, divorced and non-divorced individuals may differ systematically before divorce occurs. We therefore use 1:1 matching without replacement to construct a comparison group of individuals who are similar with respect to observed pre-treatment characteristics. The causal interpretation relies on conditional exchangeability (Rosenbaum and Rubin, 1983). This means that, conditional on the observed characteristics used in matching, individuals who divorce and comparable individuals who do not divorce are assumed to have had similar potential health trajectories in the absence of divorce. In other words, it is assumed that there are no remaining unobserved factors that jointly affect the probability of divorce and the subsequent health outcomes. However, the administrative data may not fully capture factors such as relationship conflict or social support.

We define treatment as the first observed divorce from the earliest observed marriage spell. We limit the treatment cohorts to divorces observed between 2012 and 2018 to ensure that each cohort can be observed for at least five years after divorce. The year of divorce is fixed as t_0. For each treatment year, we form a separate within-year risk set. For treated individuals whose first observed divorce occurs in t_0, eligible controls are individuals who have not yet experienced a divorce by that year. Controls may divorce later. We do not restrict the control group to individuals who never divorce because this would define control eligibility using information that occurs after t_0. Not yet divorced means that no divorce from the VA has been recorded in the panel by t_0. Since the panel starts in 2011, to avoid individual whose divorces occured before 2011 entering the control pool, we exclude controls with a recorded end of a first marriage in or before t_0. Among the matched controls, 8.4% (10,717 out of 127,395) have marital status coded as divorced in $t_0 - 1$, 6.4% are coded as married, and 83.3% have no information on marital status. Individuals must be alive at t_0 and must have an observed record in $t_0 - 1$. We exclude individuals with a musculoskeletal rehabilitation up to and including t_0. They may previously have received rehabilitation for other conditions. The resulting sample therefore consists of individuals who are at risk of their first observed musculoskeletal rehabilitation after t_0.

We require treated and control individuals to be matched exactly on `gender`, five-year `age_band` and whether any rehabilitation has occurred by $t_0 - 1$, i.e. `ever_had_any_rehab_to_date`. Then, we estimate a logistic propensity score model separately for each treatment year. The propensity score represents the estimated probability of divorce in t_0, conditional on observed characteristics measured before divorce at $t_0 - 1$. The numeric characteristics we include are `age`, `income`, `pension_income`, `cumulative_rehabs_to_date`, `cumulative_mental_rehabs_to_date`, `years_since_last_rehab`, `full_time_contribution_periods`[13], `part_time_contribution_periods`[14], `regular_employment`[15], `unemployment`[16], `marginal_employment`[17], `care_leave`[18] and `number_of_children`. The binary characteristics we include are `ever_had_mental_rehab_to_date`,

[13]Periods during which pension contributions were recorded for full-time work.
[14]Periods during which pension contributions were recorded for part-time work
[15]Days in reporting year spent in regular employment
[16]Days in reporting year spent unemployed
[17]Days in reporting year for mini- or midi-jobs
[18]Days in reporting year spent in caring capacity, *Pflegezeit*

unsuccessful_rehab_application_this_year, missing_full_time_contribution, missing_part_time_contribution and child_born_in_year. The categorical variables included are federal_state, broad_occupational_category[19], income_source[20], education[21] and youngest_child_age_group[22].

The numeric variables are standardized. Missing numeric and binary values are filled with the median and most common value respectively, along with an indicator for missingness. Categories with fewer than ten observations in a treatment year are pooled together in a separate category for 'rare' categories. We also deliberately leave marital status out of the matching. Using marital status would select many future divorcees as controls since the marital status is partially constructed using the VA which only records divorces. In an earlier run with a 4% sample, 32% of individuals who were selected as controls were eventually divorced by 2018.

We treat missingness in earnings, pension income and contribution periods as potentially informative. In particular, observed earnings with missing pension income generally indicate active employment, whereas observed pension income with missing earnings generally indicates pension receipt. We retain the observed earnings and pension values and include a categorical measure describing the joint observability of the two income sources. We also include separate indicators for missing contribution-period information. This allows the matching model to use both the recorded values and the information contained in their missingness patterns.

Within each treatment year and exact-match group, we use optimal matching (Rosenbaum, 1989). The matching procedure first maximizes the number of treated individuals who can be matched within the caliper (maximum absolute propensity score difference of 0.01) and then chooses the combination of pairs that minimizes the total propensity score distance. Since treatment years are matched in chronological order, an individual who was used as a control in an earlier year is removed from the pool of controls of later years. This individual, however, can still enter a later risk set as a treated individual if they experienced divorce. To accommodate the large annual control pools, the propensity score model is fitted using all treated individuals and, where necessary, a reproducibly selected sample of controls, with the total estimation sample limited to 500,000 observations[23]. The resulting propensity scores are corrected for the control-sampling fraction by adding the logarithm of the fraction to each individual's predicted log-odds (King and Zeng, 2001), and the fitted model is then used to calculate propensity scores for every eligible treated individual and control. The computational limit therefore affects how the propensity model is estimated but does not remove controls from the matching pool. Similarly, the optimal assignment in very large exact-match groups is solved on a reduced set of candidate pairs.

We assess covariate balance before and after matching using standardized mean

[19]Classification of Occupations 1988 created by the German Federal Employment Agency. We used the first digit which signifies occupational area.

[20]This variable has the following categories regarding the source of income: income only from working, income only from pensions, income from earnings and pensions, no income from earnings or pensions. The latter two categories are uncommon.

[21]Defined as low, medium and high where medium indicates that the highest education level is a Bachelor's degree, and high a Master's degree

[22]The age groups are 0-2, 3-5, 6-9, 10-17, 18+.

[23]Please see the section on Dealing with a large sample of controls in the Appendix

differences. Balance is assessed separately for continuous variables, binary variables, each level of categorical variables and indicators of missingness. An absolute standardized mean difference below 0.10 is treated as evidence of adequate balance on the corresponding observed characteristic (Austin, 2009). If residual imbalance above this threshold remains after matching, the affected variables are additionally included in another specification of the outcome model (see Identification section). This adjustment is intended to address limited remaining imbalance and is not used as a substitute for unsuccessful matching.

Using the matched sample, we estimate the effect of divorce on first entry into musculoskeletal rehabilitation and all-cause mortality. Musculoskeletal rehabilitation is examined over a five-year follow-up period, with death treated as a competing event. Mortality is then followed for *up to* ten years where permitted by the end of the observation period in 2023.

3.3 Identification

For identification, we focus on the average treatment effect among the matched treated individuals.

3.3.1 RQ1: Musculoskeletal rehabilitation

We follow each matched individual from t_0+1 to t_0+5 and estimate a discrete-time hazard model:

$$\begin{aligned}\lambda_{i,\tau}^{\text{rehab}} &= \Pr(R_{i\tau}=1|\text{no musculoskeletal rehab before } \tau, \text{ alive at } \tau) \\ &= \Lambda_{\text{rehab}}\left(\alpha_\tau + \beta D_i + \mathbf{x}_i'\gamma\right)\end{aligned} \tag{1}$$

where $R_{i,\tau}$ is a dummy indicating individual i's first entry into musculoskeletal rehabilitation in time in follow-up year $t_0+\tau$ where $\tau \in \{1,...,5\}$, α_τ are fixed effects, D_i is a dummy indicating divorce at t_0 and $\mathbf{x}_i$ are other variables measured at t_0-1 that we add depending on the specification. We estimate three specifications for Equation 1. In the first specification, we omit $\mathbf{x}_i$ and rely only on matching to deal with selection bias. In the second specification, we include variables whose absolute standardized mean difference after matching exceed 0.10 into $\mathbf{x}_i$. Standard errors are clustered at the level of the matched pair. In the third specification, $\mathbf{x}_i$ is instead replaced by $\mathbf{x}_{i,\tau-1}$, which is the value of each variable in the year before each follow-up year. By conditioning on variables in $t_0+\tau-1$, we remove some of the variation attributed to divorce. However, we include the specification for completeness and to identify whether other post-divorce factors can affect musculoskeletal rehabilitation. Furthermore, since matching is exact on gender, we estimate all three specifications separately for men and women.

The hazard odds ratio from Equation 1 is difficult to interpret on its own as it does not show how often rehabilitation occurs, hence we also estimate the cumulative incidence of a first musculoskeletal rehabilitation over the 5-year follow up period using a discrete Aalen-Johansen estimator[24] (Aalen and Johansen, 1978). We report four values at the the 5-year horizon, 1) cumulative rehabilitation risk among treated individuals, 2)

[24]This estimator recognizes that individuals who die can no longer enter rehabilitation. For example,

cumulative rehabilitation risk among controls, 3) risk difference per 1000 individuals and 4) the corresponding risk ratio. Confidence intervals are obtained by resampling whole matched pairs 200 times.

The biggest threat to the conditional exchangeability assumption is unobserved pre-divorce trends, which available variables may not fully capture, especially by only observing the year prior to the divorce. To address this, we first conducted an event study from $t_0 - 3$ to $t_0 + 5$ by taking the difference in values of unsuccessful rehabilitation applications, earnings, pension income, missingness of pension income, days in the year spent employed, unemployed, marginally employed, caretaking and prior rehabilitation activity between the treated and control in each matched pair and conducting a Wald test (Wald, 1943) to determine whether the differences across time are significant for each variable. We regress each within-pair difference on indivators for each year relative to t_0, with $t_0 - 1$ as the reference year and standard errors clustered by pair, and test jointly whether the coefficients for $t_0 - 3$ and $t_0 - 2$ are zero. Following divergences in trends prior to t_0, we repeated the matching exercise with three lags, i.e. $t_0 - 3, t_0 - 2, t_0 - 1$ instead of only $t_0 - 1$. Since we require that individuals have an observed record, the lag-3 specification considers only the time period between 2014 and 2018. We also repeat the lag-1 specification for only 2014 to 2018.

3.3.2 RQ2: Mortality

We follow each individual from $t_0 + 1$ until $t_0 + 10$ or 2023, the last year of the sample, whichever occurs first and estimate a discrete-time hazard model similar to Equation 1:

$$\begin{aligned} \lambda_{i,\tau}^{\text{death}} &= \Pr(Y_{i,\tau} = 1 | \text{alive at beginning of follow-up year } \tau) \\ &= \Lambda_{\text{death}}\left(\alpha_\tau + \beta D_i + \mathbf{x}_i'\gamma\right) \end{aligned} \tag{2}$$

where $Y_{i,\tau}$ is a dummy indicating that individual i dies in follow-up year $t_0+\tau$, α_τ are time fixed effects, D_i is a dummy indicating divorce at t_0 and $\mathbf{x}_i$ are other variables measured prior to t_0. Just as in Equation 1, we estimate two specifications for Equation 2 by varying the inclusion of variables in $\mathbf{x}_i$ based on absolute standardized mean differences after matching. We complement the pooled hazard odds ratio with cumulative mortality risks using the Kaplan-Meier estimator (Kaplan and Meier, 1958). We again report cumulative mortality risk among treated and controls, the risk difference per 1000 individuals and

suppose there are 100 people where 10 enter rehabilitation in $t_0 + 1$, 20 die before rehabilitation and 70 are still alive and have not entered rehabilitation. Suppose in $t_0 + 2$, 7 people out of the remaining 70 enter rehabilitation. If we treat death as censoring rather than as a competing event, the proportion surviving up to $t_0 + 1$ is $1 - \frac{10}{100} = 0.9$, and up to $t_0 + 2$ is $1 - \frac{7}{70} = 0.9$, $\therefore$ the cumulative rehabilitation probability is $1 - (0.9 \times 0.9) = 0.19$, which is an overestimation. With the Aalen-Johansen estimator, we consider the proportion which actually entered rehabilitation by each follow-up year before dying. In $t_0 + 1$, the proportion entering rehabilitation is 0.1, however, after $t_0 + 1$, the proportion remaining alive without having entered rehabilitation is $1 - \frac{10+20}{100} = 0.7$. In $t_0 + 2$, $\frac{7}{70} = 0.1$ is the share of people who enter rehabilitation, but since only 70% of the original population was alive without having entered rehabilitation at the beginning of $t_0 + 2$, the contribution to the incidence is $0.7 \times 0.1 = 0.07$, hence the cumulative rehabilitation probability is 0.17

the corresponding risk ratio at the 5-year horizon[25]. We also run the lag-3 specification with more pre-divorce years.

In addition to the straightforward analysis of divorce on mortality, we run a mediation analysis to determine how much of the mortality difference can be attributed to 'early' musculoskeletal rehabilitation as result of divorce. A single binary mediator measured one or two years after divorce is insufficient for this question since it can ignore deterioration that occurs later in the follow-up window, such as economic and rehabilitation histories that change after divorce. For example, divorce can potentially change earnings which can subsequently have the ability to affect the probability of entering musculoskeletal rehabilitation and dying.

Let $D_i \in \{0,1\}$ is divorce status fixed at t_0, $\mathbf{z}_i$ the available pre-treatment matching variables for the relevant lag-1 and lag-3 specifications, $H = 5$ the fixed mortality and mediation horizon, $\mathbf{L}_{i,t}$ the post-divorce history (`income`, `pension_income`, `unsuccessful_rehab_application_this_year`, `non_ms_rehab`[26], `income_source`, and the days in a year spent in `regular_employment`, `unemployment`, `marginal_employment` and `care_leave`) in time t and $M_{i,t-1} \in 0,1$ is a dummy indicating whether individual i entered musculoskeletal rehabilitation by the end of time $t-1$. We use information in time $t-1$ to predict information in time t. We cannot use information from time t since we are unsure whether the dependent or independent variables are recorded first. We use weighted maximum likelihood to account for individuals actually observed in the data in a given year.

We estimate the probability of an individual i having a record for the year $t_o + t$ given that they are alive:

$$\Pr(R_{i,t} = 1 | D_i, \mathbf{z}_i, \mathbf{L}_{i,t-1}) = \Lambda_{\text{obs}}(\alpha_t + \beta D_i + \mathbf{z}_i'\psi + \mathbf{L}_{i,t-1}'\psi_L + \mu M_{i,t-1}) \tag{3}$$

We then predict $\hat{p}_{obs,i}(u)$, i.e. the probability that individual i has a record in time u given their information up until $u-1$ and inverse this to compute weight:

$$\hat{\omega}_{i,t} = \min\left(\prod_{u=1}^{t} \frac{1}{\max\left(\hat{p}_{\text{obs},i}(u), \underline{p}\right)}, \bar{\omega} \right) \tag{4}$$

We set the maximum weight, $\bar{\omega} = 20$, to prevent one trajectory from having too much influence, and the minimum observation probability to $\underline{p} = 0.02$. We then estimate the parameters of a model on the probability of first musculoskeletal rehabilitation in year t given that the individual did not experience musculoskeletal rehab in $t-1$ and other information in $t-1$[27]:

[25] The hazard model uses up to ten years of follow-up although later follow-up years are available only for earlier divorce cohorts.

[26] A dummy indicating that an individual has a non-musculoskeletal rehabilitation that year.

[27] Weighed version: $\ell_{\text{rehab}} = \sum_{i,t} \hat{\omega}_{i,t} \left[M_{i,t} \log \lambda_{i,t}^{\text{rehab}} + (1 - M_{i,t}) \log \left(1 - \lambda_{i,t}^{\text{rehab}}\right) \right]$

$$\begin{aligned}\lambda_{i,t}^{\text{rehab}} &= \Pr(M_{i,t}=1|M_{i,t-1}=0, D_i, \mathbf{z}_i, \mathbf{L}_{i,t-1}) \\ &= \Lambda_{\text{rehab}}(\kappa_t + \delta D_i + \mathbf{z_i}'\eta + \mathbf{L'_{i,t-1}}\eta_L)\end{aligned} \tag{5}$$

Similarly, we estimate the parameters for a model on the probability of death in year t given that an individual was alive at the end of $t-1$[28] is given by:

$$\begin{aligned}\lambda_{i,t}^{\text{death}} &= \Pr(Y_{i,t}=1|\text{alive at the end of } t-1) \\ &= \Lambda_{\text{death}}\left(\alpha_t + \theta_D D_i + \theta_M M_{i,t} + \theta_{DM} D_i M_{i,t} + \mathbf{z}_i'\rho + \mathbf{L}'_{i,t-1}\rho_L\right)\end{aligned} \tag{6}$$

Since $\mathbf{L}_{i,t}$ is itself affected by divorce and evolves annually, we cannot hold it fixed at its observed values when generating the hypothetical divorce paths needed for the mediation decomposition[29]. Hence, we need to model the evolution of $\mathbf{L}_{i,t}$ among individuals who survive, i.e. observations where $Y_{i,t}=0$, so that we can simulate the counterfactual histories under the hypothetical divorce status for our mediation analysis (Robins, 1986; Lin et al., 2017). We fit nine separate models on the survivors, weighted by the same $\hat{\omega}_{i,t}$ (Lin et al., 2017) used in the weighted versions of Equations 5 and 6. We use a log-scale linear regressions for `income` and `pension_income`, logit for `unsuccessful_rehab_application_this_year` and `non_ms_rehab`, and multinomial logit for `income_source`. For the four employment day related counts, we use a two part model: a logit when there is at least one day and a linear regression for the number of days. These nine models condition on divorce status D_i, the covariates used for matching $\mathbf{z}_i$, $\mathbf{L}_{i,t-1}$, current mediator $M_{i,t}$, prior mediator $M_{i,t-1}$ and follow-up-year fixed effects. The two-part models for `income` and `pension_income` additionally include the current `income_source`, the models for the four employment-day counts additionally include the current `income_source`, `income`, `pension_income` and the day counts simulated before them, and the logit models include all of these, with the model for `non_ms_rehab` also including the current `unsuccessful_rehab_application_this_year`. Please see the section on Updating the evolution of $\mathbf{L}_{i,t}$ in the Appendix.

Once the parameters for these nine equations are estimated, we simulate the counterfactual histories, one year at a time using the following process.

1. Rehab probability (uses $\mathbf{L}_{i,t-1}$)→ draw→ $M_{i,t}$
    - Since our data starts from 2011, we use $\mathbf{L}_{i,2011}$ which contains true values to estimate the probability of rehabilitation, $\lambda_{i,2011}^{rehab}$. Since we need to feed $M_{i,2012}$ into the next step and not $\lambda_{i,2011}^{rehab}$, we draw a random number from a uniform distribution $U(0,1)$ and if it is smaller than $\lambda_{i,2011}^{rehab}$, then the value of $M_{i,2012}$ is set at 1 and 0 otherwise.

[28] Weighted version: $\ell_{\text{death}} = \sum_{i,t}\hat{\omega}_{i,t}\left[Y_{i,t}\log\lambda_{i,t}^{\text{death}} + (1-Y_{i,t})\log\left(1-\lambda_{i,t}^{\text{death}}\right)\right]$

[29] Suppose an individual earns 30,000 in 2013, divorces in 2014 and then earns 22,000 in 2015. Divorce could have caused the earnings to drop to 22,000 if the individual switched jobs. To determine the individual's rehabilitation or death risk if they had not divorced, we cannot plug in 22,000 because 22,000 is already corrupted by divorce. We need a clean number uncorrupted by divorce.

2. Death probability (uses $\mathbf{L}_{i,t-1}$, $M_{i,t}$) → if individual i survives, continue
3. History-update (uses $\mathbf{z}_i$, D_i, $\mathbf{L}_{i,t-1}$, $M_{i,t-1}$, $M_{i,t}$) → produces $\mathbf{L}_{i,t}$
    - We use $\mathbf{L}_{i,2011}$, $M_{i,2011}$ and $M_{i,2012}$ to simulate $\mathbf{L}_{i,2012}$.
4. Hand $\mathbf{L}_{i,t}$ forward, repeat for $t+1$

True divorce status D_i is fixed at baseline and can simply be plugged into Equation 5 for rehabilitation. In order to find the counterfactual effect, we can simply plug in the opposite value since there are only two divorce statuses and each status has a direct opposite. Hence, in step 1, we obtained the rehabilition probability and thus, predicted rehabilitation occurrence at time t, $M_{i,t}(d_i)$ for an individual with divorce status d_i. For individual i whose plugged in divorce status is $d_i = 1$, denote the rehabilitation occurrence in time t as $M_{i,t}(1)$ and when $d_i = 0$, $M_{i,t}(0)$. An individual's rehabilitation timing is given by $M_i(1)$ and $M_i(0)$[30].We then obtain the sets, $M(1)$ and $M(0)$, which contain the rehabilitation timing for all individuals.

Subsequently, when we cross $d_i \in \{0,1\}$ with the resulting $M_i(d_i)$ and $M_i(d_i')$ where $d_i \neq d_i'$, we yield four different possibilities $\in \{(0,M_i(0)),(0,M_i(1)),(1,M_i(0)),(1,M_i(1))\}$. We might be tempted to plug in these four possibilities into Equations 5, 6 and the equations for updating $\mathbf{L}_{i,t}$ (in the Appendix) but $M_i(d_i)$ is not a fixed characteristic but the output of a year-by-year stochastic process. In other words, since $M_i(d_i)$ is based on *at least one random* draw under a hypothetical divorce status, there is no direct opposite like in the case with divorce status. Furthermore, we have two dimensions, d_i and $M_i(d_i)$, where we need a counterfactual but it is difficult to defend *individual* counterfactuals for $(0,M_i(1))$ and $(1,M_i(0))$. Think of this like a natural experiment. If we only wanted a clean effect of divorce, we can externally impose divorce on some individuals. The individuals who we forced to divorce would experience the rehabilitation timing of someone with a divorced status, likewise for the non-divorced. If we wanted a clean direct effect of divorce and an indirect effect attributable to rehabilitation, we not only need to force someone to divorce but also experience the rehabilitation timing of a non-divorced person at the same time and vice versa, i.e. we need the off-diagonal or counterfactuals. This is not possible with a natural experiment as we cannot impose both dimensions simultaneously. For example, suppose we want to plug in the $(0,M_i(1))$ combination. Suppose the probability of rehabilitation under divorce is 80% and under non-divorce 5%. This means the timing of $M_i(1)$ is likely to be earlier than that of $M_i(0)$. Hence, plugging in $d_i = 0$ together with $M_i(1)$ (which is, to put it crudely, 'contaminated' with individual i's identity when $d_i = 1$) into Equation 6 would be like imposing both dimensions simultaneously. This argument also holds for the $(1,M_i(0))$ combination. To manage this 'contamination', we sever the link between individual i and their estimated rehabilitation timing by attaching another individual's rehabilitation timing. While $(0,M_i(0))$ and $(1,M_i(1))$ do not have this 'contamination' problem, we also sever the link between the individual and estimated rehabilitation timing for consistency.

To do this, we make use of the distribution of $M(1)$ and $M(0)$. This would then be similar to imposing an external rehabilitation timing. Before step 2, we replace individual i's value of $M_i(d_i)$ with individual j's $M_j(d_j)$ where $d_i = d_j$. This means that when both individuals i and j have an identical plugged in divorce status, individual i will have

[30]For example, if $M_i(d_i) = t+2$, then $M_{i,t'}(d_i) = 0$ when $0 < t' < t+2$ and $M_{i,t+2}(d_i) = 1$.

individual j's rehabilitation timing. This second-best approach helps us identify what the average death risk for individuals with a particular plugged in divorce status looks like under a random redistribution of rehabilitation timing[31].

Suppose individual i received individual j's value when $d_i = 0$, and individual k's value when $d_i = 1$. This means that individual i's hypothetical rehabilitation timings going into step 2 take the values of $M_j(0)$ and $M_k(1)$ respectively. We can then plug in the four combinations, $\in \{(0, M_j(0)), (0, M_k(1)), (1, M_j(0)), (1, M_k(1))\}$, into Equation 6 in step 2 for individual i in time t. We then update $\mathbf{L}_{i,t}$ and repeat all four steps until we reach five years.

Since individual i's $M_{i,t}$ depends on a random draw which subsequently affects $L_{i,t}$, we re-run the entire process 25 times with different draws for the rehabilitation occurrence to average out simulation randomness. Let $M_{i,r}(d_i)$ and $Y_{i,r}(d_i, M(d_i'))$ denote individual i's simulated rehabilitation and death indicators by the horizon of five years from t_0 and $R = 25$ denote the number of Monte Carlo repeats. The Monte Carlo average for individual i is given by:

$$\begin{aligned} \hat{M}_i(d_i) &= \frac{1}{R} \sum_{r=1}^{R} M_{i,r}(d_i) \\ \hat{Y}_i(d_i, M(d_i')) &= \frac{1}{R} \sum_{r=1}^{R} Y_{i,r}(d_i, M(d_i')) \end{aligned} \tag{7}$$

We then average across the matched treated individuals. Let $\mathcal{T}$ denote the set of N_M matched treated individuals:

$$\begin{aligned} \hat{M}(d) &= \frac{1}{N_M} \sum_{i \in \mathcal{T}} \hat{M}_i(d_i) \\ \hat{Y}(d, M(d')) &= \frac{1}{N_M} \sum_{i \in \mathcal{T}} \hat{Y}_i(d_i, M(d_i')) \end{aligned} \tag{8}$$

In Equation 8, we plug in the four different combinations to obtain *population* averages of the rehabilitation risk under the hypothetical divorce status, d and mortality risk under the hypothetical divorce d combined with the rehabilitation timing of divorce status d' where $d, d' \in \{0, 1\}$. We can then decompose into the direct effect and indirect effect:

$$\begin{aligned} \widehat{\text{Total effect}} &= \hat{Y}(1, M(1)) - \hat{Y}(0, M(0)) \\ \widehat{\text{Direct effect}} &= \hat{Y}(1, M(0)) - \hat{Y}(0, M(0)) \\ \widehat{\text{Indirect effect}} &= \hat{Y}(1, M(1)) - \hat{Y}(1, M(0)) \end{aligned} \tag{9}$$

[31] One caveat is that, it may be possible that individual j's rehabilitation timing is unlikely to be produced given individual i's rehabilitation probability, for example, if individual j's probability of rehabilitation in year 1 is high (e.g., 90%) while individual i's is low (e.g., 10%). This is called a positivity or overlap concern (VanderWeele et al., 2014; Petersen et al., 2012).

The total effect is simply answering what the difference in mortality risk is between divorced and non-divorced individuals within five years. The direct effect captures part of the difference in mortality risk related to divorce but unrelated to rehabilitation, i.e. given that rehabilitation timing is held constant, how does divorce (potentially operating through financial strain, loneliness, stress) affect mortality risk. The indirect effect answers how much of the difference in mortality risk is attributed to divorce shifting individuals onto the rehabilitation timing distribution typical of divorced individuals. We calculated interventional direct and indirect effects (VanderWeele et al., 2014) which is different from natural direct and indirect effects (Pearl, 2001). A natural indirect effect answers what would happen to individual i's mortality risk if their rehabilitation timing shifted from what it looks like when individual i is non-divorced to what it would look like if individual i is divorced while holding divorce status constant at $d = 1$. In contrast, an interventional indirect effect answers what would happen to average mortality risk if the population's rehabilitation timing shifts from the distribution typical of non-divorced individuals to a distribution typical of divorced individuals while holding divorce status constant at $d = 1$.

Before computing the decomposition, we check whether the fitted models reproduce the observed data. We simulate each group under its own divorce status and compare the simulated yearly cumulative incidence of musculoskeletal rehabilitation, cumulative mortality, and mean earnings and pension income with the observed values. The reason we do this is because in an earlier version of the code, due to the runaway earnings, the simulated musculoskeletal rehab ame out at 30% against the true 4.6-8.6% observed.

Finally, we then bootstrap the matched pairs 100 times (Efron and Tibshirani, 1993) with replacement and repeat the entire procedure. We do this to obtain the 95% confidence interval for the total, direct and indirect effects.

4 Results

4.1 Summary statistics

1. summary statistics table for each variable used in matching, mean, min, max

	Before matching			After matching		
Variable	Divorced	Control	SMD	Divorced	Control	SMD
Panel A: Continuous variables						
Age (years)	45.83	49.82	−0.271	45.10	45.03	0.007
Earnings (euros)	21,824	13,313	0.431	21,317	20,904	0.020
Pension income (euros)	66	213	−0.420	74	74	0.000
Full-time contribution periods	21.09	73.98	−0.415	23.62	23.02	0.007
Part-time contribution periods	3.32	8.15	−0.274	3.72	3.85	−0.008
Cumulative rehabilitations to date	0.122	0.030	0.281	0.091	0.089	0.004
Cumulative mental-health rehabilitations to date	0.053	0.008	0.223	0.037	0.034	0.014
Years since last rehabilitation[a]	1.38	1.77	−0.248	1.23	1.22	0.013
Panel B: Binary variables (%)						
Ever had a mental-health rehabilitation	4.41	0.69	0.238	3.23	2.96	0.015
Unsuccessful rehabilitation application this year	2.59	0.48	0.172	2.30	2.05	0.017
Full-time contribution data missing	91.95	73.25	0.509	90.99	91.10	−0.004
Part-time contribution data missing	91.95	73.25	0.509	90.99	91.10	−0.004
Panel C: Categorical variables (maximum |SMD| across levels)						
Variable		Levels	Before max |SMD|			After max |SMD|
Marital-status code[b]		5 / 4	3.009			0.071
Income-source status		4	0.639			0.049
Federal state		19	0.248			0.032
Occupation category		2	0.437			0.018
Age band		16 / 14	0.324			0.000
Sex		2	0.020			0.000
Any rehabilitation before $t_0 - 1$		2	0.306			0.000

Table 1: Covariate balance before and after matching

[a]Defined only for those with a prior rehabilitation; 90–98% missing, filled with a placeholder value for this comparison. [d]Before matching, divorced individuals are 100% coded as "married" by construction, while a large share of controls carry a missing marital-status code. This asymmetry inflates the pre-match SMD outside the usual ±1 range and should not be read on the same scale as the other rows.

Across all 19 variables, the mean absolute SMD falls from 0.34 before matching to 0.01 after. Earnings and pension income are set to 0 when the record for that year is unobserved. Recall that the observability of either variable is informative, reflecting current employee vs pension recipient status (`income_source`). Before matching, earnings are missing for 13.0% of treated and 41.2% of controls; after matching the missingness is 13.5% and 14.5% respectively. Pension income is missing for 91.9% of treated and 73.2% of controls before matching, and respectively 91.0% and 91.1% after matching. Among person-years with observed earnings, the mean earnings are €25,095 for treated and €22,641 for controls before matching, and respectively €24,649 and €24,446 after matching. Similarly, among person-years with observed pension points, the mean pension income is 820.15 for treated and 829.14 for controls, the mean full-time contribution period is 262.10 months for treated and 258.62 for controls and the mean part-time contribution periods is 41.23 months for treated and 43.25 for controls. The missingness rate for full-time and part-time contribution periods is not only numerically identical but matches pension income missingness almost exactly, which is consistent with all three variables tracking the same employment vs pension status. For a more detailed breakdown, please see the section on 6 in the Appendix.

4.2 RQ1: Musculoskeletal rehabilitation

	Estimate	95% CI
Panel A: Hazard model (relative scale)		
Odds ratio (treated vs. control)	1.046	[0.979, 1.117]
	$z = 1.33$,	$p = 0.184$
Panel B: Cumulative incidence at five years (absolute scale)		
Divorced	7.85%	[7.50%, 8.18%]
Controls	7.52%	[7.12%, 7.92%]
Risk difference (per 1,000)	3.2	[−1.9, 8.1]
Risk ratio	1.043	[0.975, 1.113]

Table 2: Effect of divorce on first musculoskeletal rehabilitation entry

Notes: Person-years: 219,471 Matched sets: 22,804 Events: 3,505. Standard errors clustered on the matched pair. Confidence intervals from 200 bootstrap replicates, resampling whole matched sets. Risk difference is treated minus control.

Panel A in Table 2 reports the discrete-time hazard model in Equation 1. The odds ratio is 1.046, i.e. divorced individuals have 4.6% higher odds of starting a musculoskeletal rehabilitation in any given follow-up year compared to controls. However, the estimate is not statistically significant at the 5% level. Panel B reports cumulative incidence functions on the matched cohort, with death as a competing event. 7.85% of divorce individuals have started a musculoskeletal rehabilitation compared with 7.52% of controls. This result is also not statistically significant at the 5% level.

Follow-up year	Divorced (%)	Controls (%)	Cumulative RD per 1,000	Incremental RD per 1,000	95% CI
Male					
1	1.83	1.82	0.09	0.09	
2	3.42	3.63	−2.14	−2.23	
3	5.04	5.23	−1.87	0.27	
4	6.55	6.65	−0.98	0.89	
5	7.81	7.80	0.09	1.07	[−6.4, 7.1]
Female					
1	1.50	1.63	−1.29	−1.29	
2	3.17	2.99	1.81	3.10	
3	4.83	4.42	4.05	2.24	
4	6.55	5.96	5.95	1.90	
5	7.88	7.26	6.21	0.26	[−0.4, 13.6]
Female − male, year 5			6.12		[−2.9, 16.2]

Table 3: Musculoskeletal rehabilitation: heterogeneity by gender

Notes: Cumulative incidence of first musculoskeletal rehabilitation, with death treated as a competing event. RD denotes the risk difference, calculated as divorced minus control. The sample contains 11,201 matched male pairs and 11,603 matched female pairs. Confidence intervals are reported only for the prespecified five-year endpoint. The female-minus-male estimate tests whether the five-year risk difference differs by gender ($p = 0.27$). Inference uses 200 bootstrap replications stratified by gender.

Table 3 displays the five-year cumulative effect of divorce on starting a musculoskeletal rehabilitation. For men, the five-year cumulative risk difference is 0.09 per 1,000 between divorced and controls, while for women, it is much larger at 6.2 per 1,000 although both estimates are statistically insignificant. The overall incidence among divorced men and women are also very similar at year 5 (7.81% and 7.88% respectively). The gender difference arises primarily from the contrast between divorced and controls of the same gender rather than substantially difference rehabilitation incidence between males and females overall. For women, the bulk of the risk difference is emerges in years $t_0 + 2$ to $t_0 + 4$, and the incremental difference is almost zero by year 5. For men, controls have a modestly higher cumulative incidence compared to the divorced in years $t_0 + 2$ to $t_0 + 4$. The largest difference occurs in $t_0 + 2$ where the cumulative risk among controls is 2.14 per 1,000 higher than divorced men. Similar to women, the risk difference almost disappears by year 5.

One possible interpretation is that divorce affects the transition from latent or previously untreated musculoskeletal problems into formal rehabilitation differently for men and women. Biological differences in susceptibility to musculoskeletal conditions may shape the underlying health problems on which these post-divorce processes operate, but they cannot by themselves explain the opposite divorce–control contrasts observed within gender. The timing is also consistent with differences in health-care seeking and social support following divorce. For example, the loss of spousal encouragement may delay help-seeking among some divorced men, whereas changes in household responsibilities, time constraints, or economic circumstances may alter women's willingness or ability to enter rehabilitation. These mechanisms cannot be

distinguished in the administrative data and should therefore be regarded as possible explanations rather than established channels. Moreover, the formal female-minus-male contrast is not statistically significant ($p = 0.27$), so the apparent gender difference should be interpreted as suggestive rather than conclusive evidence of effect heterogeneity.

Given these results, it is reasonable to question whether there are biological differences in susceptibility to musculoskeletal diseases. Wolf et al. (2015) find that women appear more biologically susceptible to particular degenerative and joint conditions due to differences in anatomy, hormones and biomechanics. However, certain injury patterns are more common in men, and when men develop osteoporotic hip fractures, outcomes may be worse. Differences in delayed access also

A possible interpretation is that divorce exposes the differences in approaches to healthcare seeking between genders and how those approaches interact with social support and expected household responsibilities after divorce. For example, loss of spousal support may delay care among divorced men but changes in household responsibilities may affect the willingness of women to enter rehabilitation. However, these mechanisms cannot be distinguished in the data.

DONE How much more likely are divorced people to start rehab than almost identical non-divorced people?

NA Does adding leftover covariates which were not well balanced affect the above result?

DONE Does the risk mostly show up in the year following divorce or does it creep up by year 5?

DONE What is the share of divorced people who started rehab by year 5 compared to controls?

DONE What is the difference in rehab risk between genders?

- What is the difference in the results when lag1 spec is used versus lag3 spec?

4.3 RQ2: Mortality

- How much more likely are divorced people to die during the follow up period
- Compare lag1 and lag3 spec mortality risk
- Total, direct and indirect effects plus 95% confidence interval.
- How often is it the case that the borrowed rehab timing is completely off?
- Total, direct and indirect effects for men vs women.

5 Discussion

6 Conclusion

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Appendix

Rehabilitation system in Germany

For an individual covered by statutory health insurance, musculoskeletal pain first enters the health insurance system through sickness absence rather than through the DRV. If pain makes an employee unable to work, the individual must immediately inform the employer of the incapacity and the expected duration. If the incapacity lasts longer than three calendar days, a medical certificate needs to be acquired from the individual's general practitioner, although the employer may require certification from the first day. The employer continues to pay the normal salary for up to six weeks for incapacity caused by the same illness (Bundesrepublik Deutschland, 1994). After the six weeks, if the illness continues, the statutory health insurer steps in and pays *Krankengeld*, which is usually about 70% of covered gross income up to a maximum of 90% of net income. For the same illness, this benefit is limited to 78 weeks within a 3-year period, counted from the first day of incapacity, meaning that the time of illness paid by the employer is included in the 78 weeks (Bundesrepublik Deutschland, 1988). Medical rehabilitation facilitated by the DRV requires a separate application and assessment of whether earning capacity is substantially endangered or already reduced, if rehabilitation is expected to prevent or improve that reduction (Bundesrepublik Deutschland, 1989; Deutsche Rentenversicherung, n.d.). This application is usually independent of the six weeks and *Krankengeld* although the health insurer can allow the individual ten weeks to submit a rehabilitation application indicating that earning capacity is threatened (Bundesrepublik Deutschland, 1988). For musculoskeletal conditions, the DRV assessment focuses on documented functional limitations and their consequences for work and social participation. Chronic pain, psychosocial

circumstances surrounding incapacity, comorbidity, and active cooperation are considerations in this assessment rather than four cumulative eligibility conditions. Isolated pain is insufficient to establish rehabilitation need for certain hip, knee, and shoulder conditions (Deutsche Rentenversicherung Bund, 2005). Approved medical rehabilitation is normally provided on an inpatient or full-day outpatient basis and usually lasts approximately three weeks. The employer continues wages during rehabilitation if entitlement remains, otherwise, the DRV may pay *Ubergangsgeld*, normally 68% of previous net earnings or 75% for an insured person with a qualifying child, while entitlement to *Krankengeld* is suspended (Deutsche Rentenversicherung; Bundesrepublik Deutschland, n.d. 1988).

Dealing with a large sample of controls

Suppose in 2015, there are 50,000 individuals who are divorced and 2 million eligible controls. Using all 2 million controls to estimate the logit model would cause memory issues and take a long time to run. So we imposed a cap of $450,000 (500,000 - 50,000)$ eligible controls which are then randomly selected, although the same seed is used for reproducibility. If there are less than 450,000 eligible controls, we use all of them to estimate the parameters for the logit model. Once we have estimated the logit model, we estimate the propensity score for everyone, not only the 450,000 controls and 50,000 treated. Then we match based on propensity scores and end up with a treated-control pair, i.e. in 2015, we would have 50,000 treated and 50,000 controls.

The controls used for fitting are selected by a seeded hash of the individual id, so that the same individuals are selected in every run. Since the fitting sample contains all treated individuals but only a fraction, f, of the controls, this inflates the log-odds of divorce by $\log(\frac{1}{f})$ for everyone. To rectify this, we add $\log f$ to each individual's predicted log-odds before converting it to a probability (King and Zeng, 2001).

The matching step faces a similar problem. With optimal matching, we need a distance for each treated and control pair in an exact-match group, and there can me millions of such pairs. In an earlier run on a 4% sample, divorced individuals tend to have a scores of between 0.12 to 0.19, while controls have scores around 0.001. Hence, divorced individuals may have no controls nearby. In optimal matching, these divorced individuals may be left unmatched to keep the overall distance low. We impose a 'penalty distance' which is the product of the number of unmatched individuals multiplied by 0.01. With such a penalty, then the algorithm is less likely to leave divorced individuals unmatched. If the number of pairs is too large for a dense solver, we only keep the n nearest controls where n is the number of divorced individuals in that exact-match group.

Updating the evolution of $\mathbf{L}_{i,t}$

We first estimate the parameters of the model for `income_source`. Let `income_source` be denoted by $S_{i,t}$:

$$\Pr(S_{i,t} = k|\cdot) = \frac{\exp\left(c_k + \delta_k D_i + \mathbf{z}_i'\eta_k + \mathbf{L}_{i,t-1}'\eta_{L,k} + \mu_k M_{i,t-1} + \zeta_k M_{i,t} + \tau_{k,t}\right)}{\sum_{k'} \exp(\cdot)}$$

where c_k is the constant and $\tau_{k,t}$ is the time fixed effect at year t for category k.

Then, we estimate the parameters for the models on `income` and `pension_income`. Let $A^v_{i,t}$ represent `income` and `pension_income`. The first part is a logit for whether the amount is positive:

$$\Pr\left(A^v_{i,t} \mid \cdot\right) = \Lambda\Big(a_v + b_v D_i + \mathbf{z}_i' g_v + \mathbf{L}'_{i,t-1} g_{L,v} + m_v M_{i,t-1} + n_v M_{i,t} + \tau^0_{v,t} + \phi'_v \mathbb{1}\left[\widetilde{S}_{i,t}\right]\Big)$$

The second part is a linear regression of the log amount among individuals with a positive `income` or `pension_income`:

$$\log(1 + A^v_{i,t}) = c_v + \beta_v D_i + \mathbf{z}_i' \gamma_v + \mathbf{L}'_{i,t-1}\gamma_{L,v} + \mu_v M_{i,t-1} + \zeta_v M_{i,t} + \tau_{v,t} + \phi'_v \mathbb{1}[\tilde{S}_{i,t}] + \varepsilon^v_{i,t}$$

where v indicates whether we are estimating `income` or `pension_income`, a_v and c_v are constants, $\tilde{S}_{i,t}$ was the previously estimated `income_source`, $\tau^0_{v,t}$ and $\tau_{v,t}$ are time fixed effects. In the simulation, `income` is drawn only if $\tilde{S}_{i,t}$ has the values 'earnings only' or 'earnings and pension', while `pension_income` is drawn when $\tilde{S}_{i,t}$ has the values 'pension only' or 'earnings and pension', otherwise the amount is zero. If the first part leads to a positive amount, then we add a random error within the estimated residual standard deviation. This is to introduce some variation in case the simulated individual shares identical characteristics with others. We also cap the result at the largest observed log amount.

Next, we estimate the models for the four employment days related counts $E^e_{i,t}$ where e indicates whether we are estimating the days in the year for `regular_employment`, `unemployment`, `marginal_employment` or `care_leave`. This is also the order in which we estimate and update. Each count again has two parts. The first part is a logit for $E^e_{i,t} > 0$:

$$\Pr\left(E^e_{i,t} > 0 \mid \cdot\right) = \Lambda\Big(a_e + b_e D_i + \mathbf{z}_i' g_e + \mathbf{L}'_{i,t-1} g_{L,e} + m_e M_{i,t-1} + n_e M_{i,t} + \tau^0_{e,t} + \phi'_e \mathbb{1}\left[\widetilde{S}_{i,t}\right] + \tilde{\mathbf{A}}'_{i,t}\pi^0_e + \tilde{\mathbf{E}}^{<e'}_{i,t}\theta^0_e\Big)$$

where $\tilde{\mathbf{A}}_{i,t}$ contains $\log(1 + \tilde{A}^v_{i,t})$ for the simulated `income` and `pension_income` and $\tilde{\mathbf{E}}'^{<e}_{i,t}$ contains $\log(1 + \tilde{E}^{e'}_{i,t})$ for the day counts simulated before e. For the second part, we transform the number of days into a share of the largest observed count $\bar{E}^e \le 366$, $q^e_{i,t} = (E^e_{i,t} - 0.5)/\bar{E}^e \in (0,1)$, where subtracting half a day keeps the share strictly between 0 and 1. Among individuals with at least one day, we estimate:

$$\log\left(\frac{q^e_{i,t}}{1 - q^e_{i,t}}\right) = c_e + \beta_e D_i + \mathbf{z}_i'\gamma_e + \mathbf{L}'_{i,t-1}\gamma_{L,e} + \mu_e M_{i,t-1} + \zeta_e M_{i,t} + \tau^0_{e,t} + \phi'_e \mathbb{1}\left[\widetilde{S}_{i,t}\right] + \tilde{\mathbf{A}}'_{i,t}\pi^0_e + \tilde{\mathbf{E}}^{<e'}_{i,t}\theta_e + \epsilon^e_{i,t}$$

as a first step in avoiding impossible values. We then obtain the mean $\hat{\mu}^e_{i,t}$ and residual standard deviation, $\hat{\sigma}_e = \left(\Sigma_{i,t}\hat{\omega}_{i,t}(\hat{\epsilon}^e_{i,t})^2/\Sigma_{i,t}\hat{\omega}_{i,t}\right)^{0.5}$. The number of days is then:

$$\tilde{E}^e_{i,t} = \min\left\{\max\left\{\text{round}\left(\bar{E}^e\,\Lambda\left(\hat{\mu}^e_{i,t} + \hat{\sigma}_e u^e_{i,t}\right) + 0.5\right), 1\right\}, \bar{E}^e\right\}, \qquad u^e_{i,t} \sim N(0,1)$$

We finally estimate the parameters for the model on the binary indicators `non_ms_rehab` and `unsuccessful_rehab_application_this_year`:

$$\begin{aligned}\Pr\left(B^1_{i,t} = 1 \mid \cdot\right) = \Lambda\Big(&\alpha_1 + \theta_1 D_i + \mathbf{z}'_i\rho_1 + \mathbf{L}'_{i,t-1}\rho_{L,1} + \mu_1 M_{i,t-1} + \nu_1 M_{i,t} \\ &+ \tau_{1,t} + \widetilde{\mathbf{A}}'_{i,t}\pi_1 + \mathbb{1}\left[\widetilde{S}_{i,t} = 1\right]\pi_{S,1}\Big)\end{aligned}$$

$$\begin{aligned}\Pr\left(B^2_{i,t} = 1 \mid \cdot,\, B^1_{i,t}\right) = \Lambda\Big(&\alpha_2 + \theta_2 D_i + \mathbf{z}'_i\rho_2 + \mathbf{L}'_{i,t-1}\rho_{L,2} + \mu_2 M_{i,t-1} + \nu_2 M_{i,t} \\ &+ \tau_{2,t} + \widetilde{\mathbf{A}}'_{i,t}\pi_2 + \mathbb{1}\left[\widetilde{S}_{i,t}\right]\pi_{S,2} + \pi_B \widetilde{B}^1_{i,t}\Big)\end{aligned}$$

where $B^1_{i,t}$ is a dummy representing `unsuccessful_rehab_application_this_year` and is estimated first, $B^2_{i,t}$ is a a dummy representing `non_ms_rehab` which is estimated second hence also contains the predicted $\widetilde{B}^1_{i,t}$ and $\widetilde{\mathbf{A}}_{i,t}$ is a vector containing the drawn level ($\log(1+x)$ not log) of the simulated current amounts of (`income`, `pension_income`, and days in `employment`, `unemployment`, `marginal_employment` and `care_leave`) which we just estimated in the regressions above.

Descriptive summary

Year	Person-years	Individuals	Divorces	% of panel	All rehab	MSK rehab	Deaths
2011	2,858,290	2,858,290	4,731	0.166	43,880	19,051	6,310
2012	2,867,809	2,867,809	4,496	0.157	45,951	20,084	6,457
2013	2,879,401	2,879,401	4,393	0.153	45,836	21,610	6,256
2014	2,900,975	2,900,975	4,239	0.146	46,961	22,618	6,412
2015	2,926,410	2,926,410	4,021	0.137	47,158	22,501	5,871
2016	2,954,800	2,954,800	3,804	0.129	45,724	21,585	6,595
2017	2,981,776	2,981,776	3,655	0.123	45,194	21,184	6,568
2018	3,022,330	3,022,330	3,493	0.116	46,460	21,531	7,360
2019	3,050,013	3,050,013	3,283	0.108	48,882	22,958	7,251
2020	3,054,308	3,054,308	3,040	0.100	39,346	16,985	8,769
2021	3,068,487	3,068,487	2,714	0.088	42,638	2,407	12,130
2022	3,105,388	3,105,388	2,302	0.074	44,671	16	12,905
2023	3,128,428	3,128,428	1,858	0.059	45,196	0	13,972
2024	3,146,317	3,146,317	—	—	—	—	10,614
Total	41,944,732	3,717,941	46,029		587,897	212,530	

Table 4: Overall data description

Notes: Each individual is counted once per year and once in the total, so the total is not the sum of the column. Divorces are counted in the year of the first divorce we observe. Of the 46,029 individuals, 45,330 divorce out of the earliest marriage we observe. Rehabilitation counts are successful, completed rehabilitations. MSK stands for musculoskeletal.

The death column counts records whose year matches the recorded year of death. It is descriptive only. The mortality analysis uses the recorded year of death directly, not these annual records. The 2024 records carry no divorce or rehabilitation information. Delete the 2024 row. To do: create a new overall data description table where deaths are counted correctly. Rehab codes changed in 2021, musculoskeletal rehab from 2021 needs to include the new codes 53, 56 in addition to leftover code 20. Rehab codes from 2021 range from 40 to 59.

The number of divorces fell steadily from 4,731 divorces in 2011 (0.166% of individuals observed in 2011) to 1,858 divorces in 2023 (0.059% of individuals in 2023). The national divorce rate also fell during this period although at a slower rate. The reason for this difference may be attributed to who gets counted in the VA dataset. The VA dataset explicitly excludes marriages which lasted three years or less or if the individuals divorcing waived the pension equalization through a prenuptial agreement or divorce settlement. Furthermore, if the income of the individuals divorcing are roughly equal, then the family court allows pension equalization to be skipped (Deutscher Bundestag, 2009; Koch, 2025).

	Category	Codes	Events	Unique persons	Share (%)
1	Musculoskeletal[a]	20/53, 56	273,543	≤174,646	46.5
2	Mental health[a]	30/55	89,711	≤72,632	15.3
3	Neoplasms[b]	26/48	71,795	≤56,453	12.2
4	Nervous system[b]	28/52	31,133	≤22,350	5.3
5	Cardiovascular[c]	22	18,000	14,979	3.1
6	Digestive & metabolic[c]	23	16,469	13,737	2.8
7	Skin[b]	25/57	15,821	≤12,457	2.7
8	Other conditions[c]	31	13,649	10,627	2.3
9	Endocrine[c]	24	11,743	9,818	2.0
10	Respiratory[c]	21	7,999	6,819	1.4
	Top-10 total		549,863		93.5
	All groups		587,897	350,935	100.0

Table 5: Top ten rehabilitation diagnosis groups

Notes: The DRV reclassified rehabilitation approvals from ICD-10-based diagnosis groups (codes 20–33) to treatment-based indicator groups (codes 40–59) on 1 January 2021. Combined groups show old/new codes; event counts are summed across both systems. [a]Mapping confirmed in DRV documentation. [b]Mapping inferred from year-by-year volume patterns; code 52 (neurology) may additionally include stroke cases previously under code 22 (cardiovascular). [c]Old-system code only since it is possible that the indicator groups system redistributes these diagnostic chapters across multiple treatment-based codes (principally code 51, internal medicine), preventing one-to-one mapping; counts cover 2011–2021 only. Unique-person counts for combined groups are upper bounds (≤) because the same individual may appear under both coding systems during the 2020–2021 transition.

Across the 4% sample, 350,935 individuals have at least one completed rehabilitation (about 9.4% of all individuals in the sample). There are 587,897 rehabilitations covered. In any given year, about 1.3% of the observed population starts a rehabilitation. The most common rehabilitation type is for musculoskeletal related conditions, covering 46.5% off all rehabilitations, followed mental health rehabilitation, covering 15.3% of all rehabilitations.

	Treated		Controls	
Restriction	Kept	Excluded	Kept	Excluded
First divorce in 2012–2018	28,101		20,533,501	
Not yet divorced at t_0	—	—	20,384,393	149,108
Divorce ends first-observed marriage	27,891	210	—	—
Alive at t_0	27,887	4	20,317,221	67,172
No MSK rehab starting in t_0	27,343	544	20,201,989	115,232
Has a record in $t_0 - 1$	27,202	141	19,697,995	503,994
Has a valid age in $t_0 - 1$	27,202	0	19,697,928	67
No MSK rehab before t_0	25,541	1,661	19,334,802	363,126
Eligible	25,541	2,560	19,334,802	1,198,699

Table 6: lag-1 specification sample, 2012 to 2018

Notes: Treated individuals enter in exactly one year, so the treated columns count people. Controls can be eligible in several years, so the control columns count year slots rather than people. Hence, the 19,334,802 eligible control slots belong to 3,156,224 individuals.

To build the matches for the lag-1 specification, we use first observed divorces from 2012 to 2018 so that each cohort has at least five years of follow-up. In this window, there are 28,101 divorces. After imposing restrictions, we are left with 25,541 individuals. See Table 6 above.

t_0	Eligible treated	Eligible controls	Matched pairs	Match rate (%)	Average gap in a pair	Average score treated	Average score controls
2012	4,317	2,765,104	4,290	99.4	0.00008	0.118	0.0014
2013	4,142	2,754,989	4,110	99.2	0.00019	0.128	0.0013
2014	3,912	2,749,187	3,870	98.9	0.00027	0.132	0.0012
2015	3,679	2,750,172	3,608	98.1	0.00040	0.143	0.0012
2016	3,359	2,756,249	3,142	93.5	0.00069	0.152	0.0010
2017	3,164	2,770,496	2,254	71.2	0.00081	0.169	0.0010
2018	2,968	2,788,605	1,530	51.5	0.00098	0.186	0.0009
Total	25,541		22,804	89.3	0.00040		

Table 7: Matching by treatment year

Notes: Eligible controls are year slots available before matching. The gap in a pair is the absolute difference in propensity scores.

Table 7 shows the matching results. Of the 25,541 eligible treated individuals, we managed to match 22,804 (89.3% of all treated) and fail to match 2,737 (10.7% of all treated). The matched sample is therefore 45,608 individuals and no control is used twice. Every treated individual has at least one control in their exact-match group (i.e. in their gender, five-year age band and prior rehabilitation present group). The matched sample has close to an even split by gender. Across all years the median absolute gap in propensity scores is is 0.00003, the 95th percentile is 0.0022 and the largest gap is 0.0099.

The match rate is nearly complete from 2012 to 2015 and then decreases. It is likely because controls used in earlier years are taken out of later pools. Furthermore, the pool

of people who have not yet divorced shrinks over time. Why? The individuals we matched have a mean age of 45.1 while the unmatched individuals are 51.9 on average. Of the matched treated individuals, 7.3% have had some rehabilitation before t_0 compared to 28.0% of unmatched treated individuals. 86% of unmatched individuals come from 2017 and 2018. Hence, our subsequent results apply to divorced individuals who tend to be younger and healthier. It is also possible for individuals to appear first as a control and then as a treated if that individual divorces eventually. 7,273 individuals out of 38,335 individuals across all matched pairs are first controls and then treated (about 19.0%). There is a mean gap of two years between being a control and being a treated.

Table 8: Differences between matched pairs from t_0-3 to t_0+5

Year relative to t_0	Earnings (euros)	Pension income (euros)	Rehabilitation starts (pp)	Unsuccessful applications (pp)
-3	-586.9	3.33	-0.11	-0.04
	(124.4)	(1.25)	(0.24)	(0.19)
-2	-417.2	1.18	0.21	0.28
	(80.6)	(0.91)	(0.24)	(0.18)
-1	0	0	0	0
	—	—	—	—
0	32.6	-0.86	0.66	-0.16
	(57.8)	(0.75)	(0.22)	(0.17)
1	-34.5	0.30	-0.35	0.05
	(84.3)	(1.17)	(0.27)	(0.18)
2	-337.8	2.14	0.09	-0.20
	(101.1)	(1.43)	(0.28)	(0.18)
3	-516.9	2.04	-0.11	-0.22
	(114.1)	(1.64)	(0.29)	(0.19)
4	-714.0	1.49	-0.32	-0.51
	(125.6)	(1.93)	(0.29)	(0.19)
5	-854.8	2.91	0.14	-0.36
	(137.4)	(2.22)	(0.30)	(0.19)
Test statistic	30.65	7.04	1.49	3.51
p-value	< 0.001	0.030	0.474	0.173

Notes: Standard errors in brackets are clustered on the matched pair. The last two rows test whether the differences at t_0-3 and t_0-2 are both zero. The number of pairs is 14,127 at t_0-3, 18,287 at t_0-2, 22,804 at t_0-1 and t_0, and 22,112 at t_0+5. The t_0-3 row uses only the 2014 to 2018 cohorts, because the panel starts in 2011. Earnings and pension income in t_0-1 are matching variables, so the difference is 0 by construction.

In the lag-1 specification, matching only makes the treated and controls similar in t_0-1. It does not guarantee that both groups were on a similar path before t_0. Divorced individuals are seen having a different earnings path before t_0-1 already. This suggests that the lag-1 specification may not satisfy matching assumptions, hence we also report the lag-3 specification.

	Divorced		Controls	
What happened	n	%	n	%
Neither rehabilitation nor death	20,558	90.15	20,811	91.26
Died before any MSK rehabilitation	467	2.05	285	1.25
Started MSK rehabilitation, survived	1,770	7.76	1,701	7.46
Started MSK rehabilitation, then died	9	0.04	7	0.03
Total	22,804	100	22,804	100

Table 9: Outcomes in the five years after t_0

Notes: Matched sample, lag-1 specification, five years from t_0. MSK stands for musculoskeletal. Deaths come from the recorded year of death. One individual has a rehabilitation and a death recorded in the same year. Over the longer mortality window of up to ten years, the matched sample records 2,017 deaths.

In the five years after t_0, most individuals (90.2% among treated and 91.3% among controls) neither experience musculoskeletal rehabilitation or death. 7.8% of the treated and 7.5% of controls have started musculoskeletal rehabilitaion. 2.05% of the treated die without starting musculoskeletal rehabilitation compared to 1.25% of controls. Note that not everyone has a record in every follow-up year. 878 individuals out of the 38,335 (2.3%) are missing at least one yearly record within the five-year window (1,282 missing records out of 219,471). We keep the full five-year window and treat a missing year as a year with no record rehabilitation instead of dropping the individual. A missing year might indicate labor market exit as a result of poor health, so we also run a stricter version where both members of a pair stop being followed just before the first year either one of them is missing a record. Since this affects 909 pairs where the mean gap is in $t_0 + 3$, the sample falls to 22,612 pairs, while the number of musculoskeletal rehabilitations falls from 3,505 to 3,449.

DONE How many individuals are in the 20% dataset? How many person years?

DONE How many individuals have undergone divorce each year?

DONE Matching limited to 2012 to 2018 for lag1 spec, and 2014 to 2018 for lag3 spec. Who do we filter out?

DONE How many people undergo medical rehabilitation?

DONE What are the top rehabilitation codes and cases?

DONE Sample size and number of matches used for each. What is the matching rate by treatment year?

DONE N_T: eligible treated individuals, N_M: matched individuals (treated and controls), gender composition. Are there people who show up twice, perhaps first as control and then as divorced down the line?

DONE Distribution of propensity scores, mean, max between matched individuals

DONE Event study coefficients, pre-trend Wald test/joint coefficient test

DONE Do people have gaps in the data?

DONE What happens in the five year follow up? Do people die before they get the rehab? How many complete the rehab and survive the 5 years? How many neither get rehab nor die in the 5 years?

DONE SMD differences